



**PSG INSTITUTE OF TECHNOLOGY
AND APPLIED RESEARCH COIMBATORE - 641 062
(Autonomous)**

Final Year Project

ClinWise - AI Chatbot for Trial Completion Prediction

“Helping researchers forecast trial outcomes and reduce design risks.”

Team Members:

1. Padma Priya R (715522244032)
2. Sanjeevi V (715522244044)
3. Tarunaa A C (715522244054)

Project Guide:

Dr. Manimegalai

Agenda



- Introduction
- Scope and Motivation
- Literature Survey and Gaps identified
- Problem Statement and Objectives
- Proposed methodology: AI Chatbot - Trial Completion Prediction
- Hardware and Software Requirements
- Roles and Responsibilities and Project Timeline
- Conclusions
- References



Introduction

- A clinical trial is a research study conducted on human participants to evaluate the safety, efficacy, and effectiveness of medical interventions.
- They are the gold standard for developing new drugs, vaccines, and therapies.
- Conducted in multiple phases (I–IV) to progressively test safety, dosage, side effects, and long-term effectiveness.
- Outcomes of trials directly influence regulatory approvals, healthcare practices, and patient safety.



Issues and Challenges in Clinical Trial

- **High failure rate:** Many trials fail due to poor design, unrealistic recruitment goals, or limited patient eligibility.
- **High cost & time-intensive:** Clinical trials require large investments, often extending over several years.
- **Complexity of protocols:** Lengthy eligibility criteria and strict regulations limit patient participation.
- **Lack of predictive insight:** Difficulty in forecasting trial success before initiation.
- **Underutilization of data:** Large volumes of trial-related data remain unstructured and unused effectively.



Scope & Motivation

- Growing demand for faster and cost-effective drug development.
- Increasing availability of open-access trial datasets provides an opportunity for advanced analytics.
- Traditional methods lack the ability to combine structured fields with textual descriptions for holistic analysis.
- Project addresses these gaps by integrating predictive modeling, explainability, and user-friendly deployment (e.g., dashboards or chatbot).

Literature Survey



S.No	Details of the paper	Problem solved	Proposed solution	Results achieved	Merits & Demerits/Future work
1	Ece Kavalci & Anthony Hartshorn, “Improving clinical trial design using interpretable machine learning based prediction of early trial termination,” <i>Scientific Reports</i> , 2023.	Early termination of trials wastes resources.	Used structured + eligibility data with ML (RF, XGBoost) and SHAP for interpretability.	Achieved moderate prediction accuracy; SHAP revealed top delay drivers (phase, site count, sponsor).	Merits: Large dataset, interpretable. Demerits: Needs external validation, deeper protocol features.
2	L. Qian et al., “Enhancing clinical trial outcome prediction with AI,” <i>ScienceDirect</i> , 2025.	Lack of advanced AI integration in trial prediction.	Reviewed embedding & multimodal learning for trial outcomes.	- ML model outperformed linear and industry baseline methods. - More accurate ranking of sites based on expected recruitment.	Merits: Good overview, identifies methods. Demerits: Limited benchmarking, calls for unified evaluation.

Literature Survey



S.No	Details of the paper	Problem solved	Proposed solution	Results achieved	Merits & Demerits/Future work
3	A. Aliper et al., “Prediction of Clinical Trials Outcomes Based on Target Choice,” inClinico, 2023.	Difficulty predicting Phase II outcomes.	Transformer-based AI using drug target features.	Improved accuracy vs baselines.	Merits: Modern deep learning, early phase focus. Demerits: Limited interpretability, more diverse data needed.
4	Z. Wang, C. Xiao, J. Sun, “SPOT: Sequential Predictive Modeling of Clinical Trial Outcome,” arXiv, 2023.	Existing models ignore trial similarity & time.	Meta-learning with sequential trial embeddings.	Predictive accuracy moderate; identified top influencing factors (phase, sponsor).	Merits: Better accuracy, handles time trends. Demerits: Complex, less interpretable.

Literature Survey



S.No	Details of the paper	Problem solved	Proposed solution	Results achieved	Merits & Demerits/Future work
5	Ling Yue et al., “TrialEnroll: Predicting Clinical Trial Enrollment Success,” arXiv, 2024.	Enrollment failure leads to incomplete trials.	Deep & Cross Network + LLM text features.	PR-AUC 0.70, better than baselines.	Merits: Focuses on enrollment, interpretable text. Demerits: Needs validation across diseases.
6	J. Cho et al., “Predicting clinical trial duration via ML,” <i>PMC</i> , 2025.	Hard to estimate trial duration.	Survival models + ML using trial features.	Reasonable predictive power for duration.	Merits: Combines ML + statistics. Demerits: Duration ≠ completion, more unstructured data needed.

Literature Survey



S.No	Details of the paper	Problem solved	Proposed solution	Results achieved	Merits & Demerits/Future work
7	R. Alkhanbouli et al., “The role of explainable AI in disease prediction,” <i>PMC</i> , 2025.	Lack of trust in black-box AI.	Review of XAI methods in health.	Identified SHAP, LIME, attention methods.	Merits: Highlights XAI benefits. Demerits: Few works on trials, need practical adoption.
8	D. Teodoro et al., “AI in Clinical Trials Risk Assessment,” <i>NPJ Digital Medicine</i> , 2025.	Limited studies on AI risk prediction in trials.	Scoping review of risk types & methods.	Found more in operational risk; fewer in efficacy.	Merits: Good overview, maps gaps. Demerits: Need standard datasets, more efficacy focus.

Gaps identified

1. Narrow Focus on Single Aspects

- Existing works focus on isolated factors such as trial termination, enrollment, or duration.
- Very few adopt a holistic approach that combines multiple risk factors together.

2. Underutilization of Text Data

- Majority of models rely mainly on structured data.
- Rich unstructured data such as eligibility criteria, protocol descriptions, and intervention texts remain largely untapped.

Gaps identified

3. Lack of Model Interpretability

- Deep learning models achieve higher accuracy but provide little interpretability. This limits adoption by regulators, clinicians, and sponsors who need transparent and explainable outputs.

4. Limited Use of Advanced Embeddings & Multimodal Learning

- Potential of embeddings and multimodal learning (drug features + text + sponsor profile) is underexplored.
- Actual implementations remain rare despite being highlighted in recent reviews.



Problem Statement

- Clinical trials are the backbone of medical innovation, validating the safety and efficacy of new drugs and treatments.
- However, the process is long, expensive, and uncertain, often spanning several years. A large number of trials fail to complete successfully, leading to:
 - Substantial financial losses for sponsors and organizations
 - Delayed patient access to potentially life-saving therapies
 - Wastage of valuable research resources

Key Objectives

- Build an explainable AI model that predicts whether a clinical trial will be Completed or Not Completed by combining structured metadata with unstructured text (titles, summaries, and eligibility criteria) for higher accuracy.
- Deploy the model as a chatbot interface that provides outcome predictions along with SHAP-based explanations.
- Provide actionable recommendations to support early decision-making.

The Proposed Methodology: AI Chatbot - Trial Completion Prediction



Data Collection

- **Primary Source:** ClinicalTrials.gov – one of the largest publicly available repositories of clinical trial records.
- **Dataset Size:** ~450,000 trials, covering a wide range of disease areas, interventions, and trial phases.
- **Metadata Captured:** Trial identifiers (NCT number), status, phase, enrollment, sponsor, funder type, study design, conditions, interventions, outcomes, Study title, brief summaries, eligibility criteria, outcome descriptions.

Rich combination of structured and unstructured features allows a more holistic prediction of trial completion likelihood.

Data Cleaning



- 1.Remove irrelevant columns
- 2.Handing missing values
- 3.Date conversion
- 4.Target variable creation
- 5.Filtering to industry funded trials

Feature Engineering

Structured Features: Encode categorical variables using one-hot or target encoding and scale numerical features as appropriate.

Derived Features and Interactions:

- Interaction between study design and funder type, such as randomised controlled trial with industry funding.
- Interaction between phase and sponsor, such as phase one trials sponsored by specific organizations.
- Multiplicative interaction of enrollment and phase.

Feature Engineering

1. Phase Number Extraction

“Phase 1”, “Phase 2” → numerical mapping → phase_num

2. Duration per phase

$\text{duration_per_phase} = \text{Duration_days} / \text{phase_num}$

3. Risk score extraction("terminated","withdrawn","safety","funding","slow recruitment")

4. Sponsor success rate

Model development

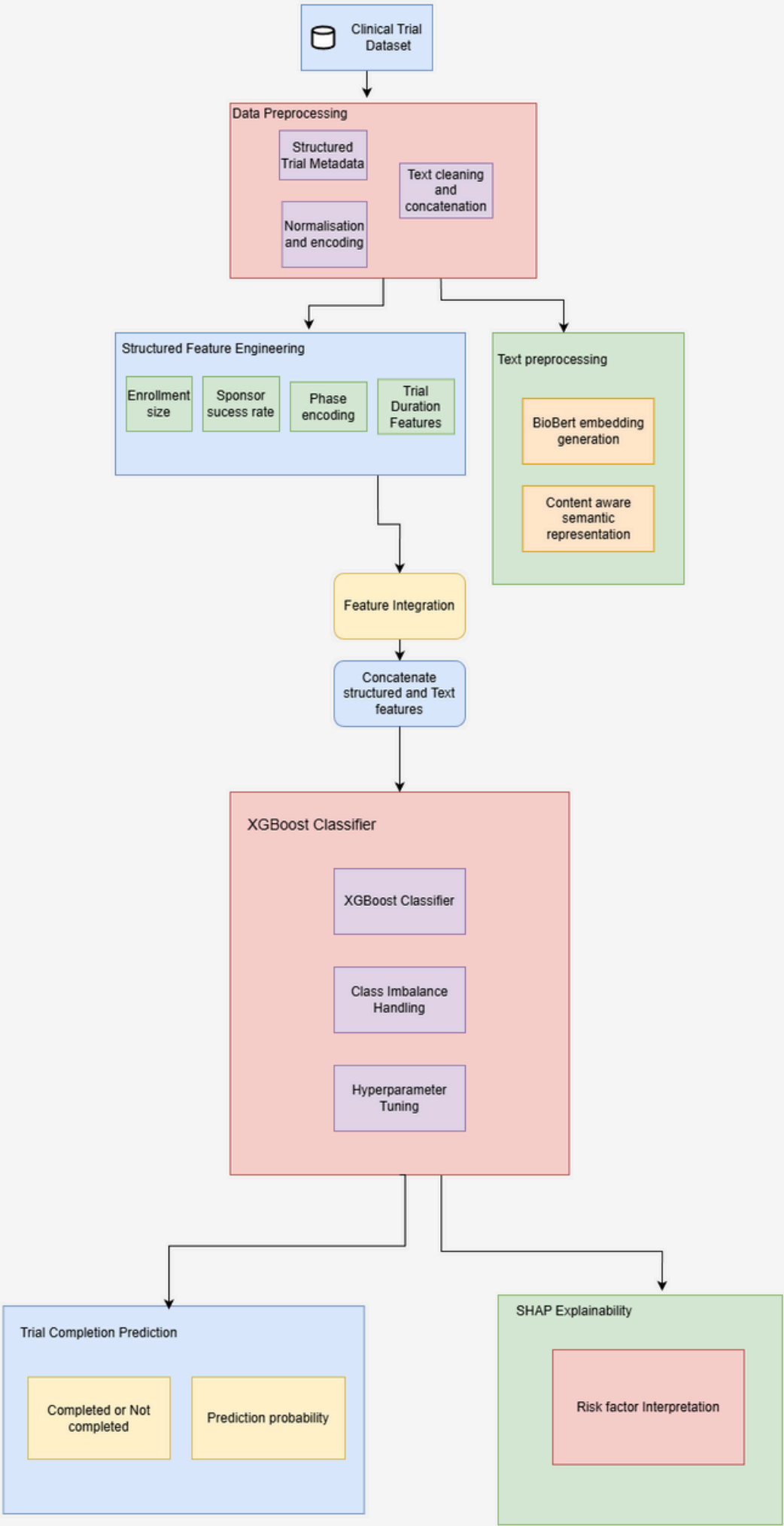
- Combine structured and unstructured features for a multimodal predictive model.
- **Unstructured Data Models:** Biobert for text embeddings.
- **Multimodal Integration:** Concatenate structured features with text embeddings to train fully connected layers.
- **Explainability:** SHAP for feature-level impact on individual predictions.
- **Evaluation Metrics:** Precision, Recall, F1-Score, AUC-ROC, Confusion Matrix.

Chatbot Decision Support

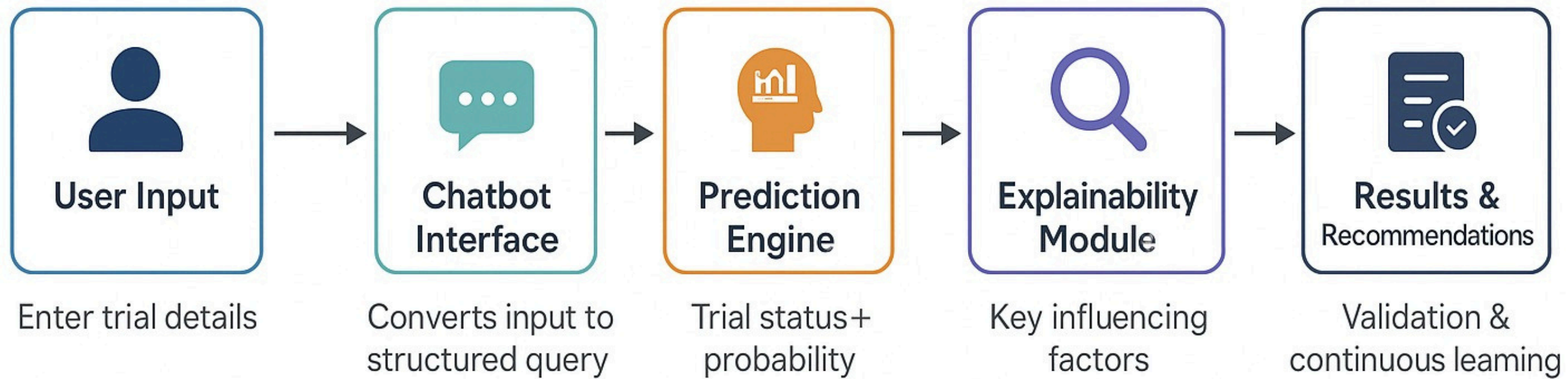


- Interactive system implemented using React
- Conversational interface presenting predictions in natural language
- Explanation of key risk factors influencing trial completion
- Generation of actionable recommendations through counterfactual analysis

Architecture Diagram



Workflow of ClinWise



Model Performance comparision

Model	Accuracy	F1-Score	ROC-AUC
Logestic Regression	0.61	0.67	0.7057
Random Forest	0.66	0.71	0.7463
LightGBM	0.87	0.87	0.8366
XGBoost(Final Model)	0.87	0.92	0.9087

Hardware requirements

- **Development PC:** 16 GB RAM, GPU-enabled recommended
- **Storage:** 50 GB+ for dataset and embeddings

Software requirements

- Python (Pandas, NumPy, Scikit-learn, PyTorch/TensorFlow)
- **NLP:** Biobert embeddings for capturing medical terms
- **Explainability:** SHAP
- **Backend:** Flask
- **Frontend:** React
- **Dataset format:** CSV, Excel

Roles and Responsibilities

1.Tarunaa - Data Engineer / Scientist

- Prepare and engineer clinical trial data for modeling.

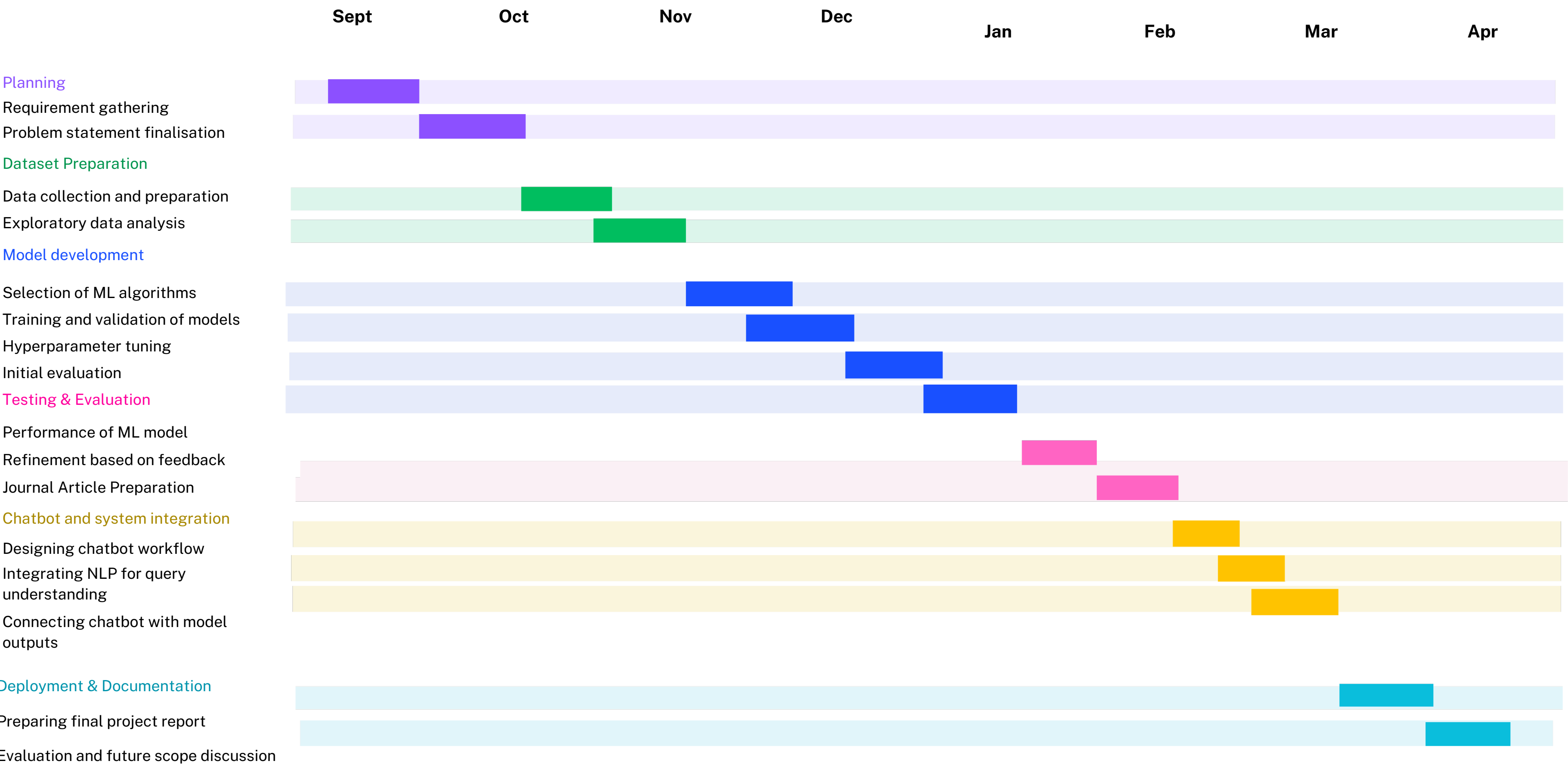
2.Padma Priya R - Machine Learning Engineer

- Build, train,test, and evaluate predictive models with explainability.

3.Sanjeevi V - Deployment & Clinical Liaison:

- Deploy the model and ensure clinical usability and interpretability.

Project Timeline 2025 - Gantt Chart



Conclusions

- Clinical trial design remains complex, with high risks of delays and failures.
- Our project addresses this challenge by predicting trial completion status using AI and deep learning.
- By integrating structured and unstructured data, the model provides not only predictions but also explainability and causal insights.
- The proposed chatbot interface makes these insights accessible in natural language, improving usability for scientists and stakeholders.
- Overall, this solution can enhance decision-making, reduce errors, and improve efficiency in clinical trial design.

Future Work

1. Voice Based Interaction

- Enable voice input so users can speak trial details instead of typing. The chatbot responds with synthesized speech for a hands-free experience.

2. Chatgpt like memory of past trials

- Store previous trial queries and predictions for long-term reference. Helps the chatbot provide continuity, comparisons, and personalized insights.

3. Prediction of operational delays

- Extend the model to estimate delays in enrollment, timelines, and completion.
- Gives organizations early warnings to manage operational risks.

4. Real Time Data Fetching from ClinicalTrials.gov

- Automatically pull updated trials, statuses, and results from the API.
- Ensures the system always uses the latest and most accurate research data.

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